34) Use the vector $u = (u_1, \ldots, u_n)$ to verify the following algebraic properties of R^n.

a) $u + (-u) = (-u) + u = 0$

b) $c(du) = (cd)u$ for all scalars c and d

Solution: a) $(-u) = (-1) \cdot u = (-1) \cdot (u_1, \ldots, u_n) = (-1 \cdot u_1, \ldots, -1 \cdot u_n)$ (Scalar multiplication)

$= (-u_1, \ldots, -u_n)$ $\therefore u + (-u) = (u_1, \ldots, u_n) + (-u_1, \ldots, -u_n)$

$= (u_1 + (-u_1), \ldots, u_n + (-u_n))$ (Vector addition)

$= ((-u_1) + u_1, \ldots, (-u_n) + u_n)$ (Algebraic property (i))

$= (0, 0, \ldots, 0)$ (Algebraic property (iv)) $= 0$

b) $du = d(u_1, \ldots, u_n) = (du_1, \ldots, du_n)$ (Scalar multiplication)

$c(du) = c(du_1, \ldots, du_n) = (c(du_1), \ldots, c(du_n))$ (Scalar multiplication)

$= ((cd)u_1, \ldots, (cd)u_n)$ (Algebraic property vii)

$= (cd)(u_1, \ldots, u_n)$ (Scalar multiplication) $= (cd)u$ $\square$